# DFS Water Jug Problem (Python)

## Introduction

This program solves the Water Jug Problem using the Depth First Search (DFS) algorithm. There are two jugs with capacities of 4 liters and 3 liters. The goal is to measure exactly 2 liters of water using different operations such as filling, emptying, and pouring water between the jugs.

## Purpose of Variables

CAP_A: Capacity of Jug A (4 liters).

CAP_B: Capacity of Jug B (3 liters).

GOAL: Target amount of water to achieve (2 liters).

## Functions Used

### get_neighbours()

This function generates all possible next states from the current state of the jugs. It includes actions like filling a jug, emptying a jug, or pouring water from one jug to another.

### dfs_water_jug()

This function implements the Depth First Search algorithm. It explores different states of the jugs step by step until the goal amount of water is reached. It also keeps track of visited states, the path taken, and the rules (operations) used.

## Working of the Program

The program starts from the initial state (0,0) where both jugs are empty. It then explores possible moves using DFS and continues searching until one of the jugs contains the target amount of water. Finally, the program prints the steps and total number of moves required to reach the solution.

## Conclusion

This program demonstrates how the DFS algorithm can be used to solve state-space problems such as the Water Jug Problem. It shows how different states are explored until the goal condition is achieved.